

SYSTEMATIC REVIEW

Deep learning applications in prosthodontics: A systematic review

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ABSTRACT

Statement of problem. Deep learning (DL) has been applied to aid dental professionals in diagnosis, treatment planning, and fabricating prostheses. However, an overview and the status of the main DL applications in prosthodontics is lacking.

Purpose. The purpose of this systematic review was to evaluate DL applications for inlays, onlays, and tooth-supported crowns and fixed dental prostheses (FDPs), by predicting restoration outcomes, optimizing prosthetic design, assisting treatment planning, improving color matching, and automating landmark detection for removable partial denture (RPD) design and facial changes after complete denture (CD) treatment.

Material and methods. A systematic review was completed in 6 databases: PubMed, EMBASE, Scopus, Web of Science, arXiv, IEEE, and Google Scholar. A manual search was also conducted. Two investigators evaluated the studies independently using the JBI Critical Appraisal checklist. A third examiner was consulted to resolve any lack of consensus. The included articles were classified based on the DL application: identification of restorations, prediction of restoration outcomes (such as debonding, wear, fracture resistance, surface roughness, microhardness, and flexural strength), assistance in treatment planning, prostheses design and manufacturing, shade matching, landmark detection for RPD planning, and analysis of facial changes after CD treatment.

Results. Of 3359 screened studies, 31 met the eligibility criteria for inclusion. Among these, 10 studies demonstrated a low risk of bias across all domains of the JBI checklist. Most of the reviewed studies were concentrated on the design and manufacturing of dental prostheses (n=13) and used a variety of deep learning applications with generation (n=11) as the predominant task. The Convolutional Neural Network (CNN) was the most utilized model, appearing in 11 studies, followed by Generative Adversarial Networks (GAN) in 7 studies. Among the restorations considered, tooth-supported crowns were the most frequently assessed (n=14). Regarding data modalities, intraoral scanners (IOSs) were the most utilized in the studies (n=16). The highest accuracy of 99.4% for identifying gold restorations and the lowest accuracy of 60% for detecting an onlay and other restoration designs were reported. Based on the included studies, sensitivity ranged from 88.6% to 100%, and Intersection over Union (IoU) ranged from 60% to 90%.

Conclusions. DL in prosthodontics, especially concerning prosthesis design and manufacturing, demonstrates significant potential. However, the standardization of methodologies and rigorous validation are essential to ensure the reliable and widespread clinical adoption of these DL-driven approaches. (J Prosthet Dent xxxx;xxx:xxx-xxx)

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Clinical Implications

Deep learning models for prosthodontics are still being developed but have the potential to serve as powerful clinical tools.

Deep learning (DL), a complex subfield of machine learning and artificial intelligence (AI), utilizes layered data-processing structures to learn from data at multiple levels of abstraction.¹ Unlike traditional machine learning, DL's deep neural networks require significantly more data to effectively learn autonomously.¹ This approach acknowledges the inherent differences between computer-based and human intelligence. DL algorithms have been engineered to assist different clinical applications, including detecting dental diseases such as caries or oral neoplasms,² identifying edentulous regions or types of dental implants from periapical radiographs and panoramic images, predicting peri-implantitis, and estimating the success rates of implant-supported prostheses.³

DL models have also been applied to prosthodontics aiming to advance the development of materials and to assist in diagnostics, treatment planning, and fabricating dental prostheses. These applications vary from designing removable partial dentures (RPDs)⁴ and detecting finish lines of tooth preparations to being a tool integrated into the design and manufacturing of fixed and removable dental prostheses.⁵ Moreover, these AI models can assist with tasks such as identifying tooth preparation boundaries and selecting tooth shapes for computer-aided design (CAD)-created dental prostheses.⁶ DL may also help with matching color, identifying the finish line of tooth preparations, or aiding in choosing tooth shapes for the design of dental prostheses using CAD software programs.^{6,7} These AI-based algorithms may also optimize the manufacturing process of castings or predict facial changes caused by removable dental prostheses.⁷

While DL in prosthodontics has improved and shown diverse results, the integration of these AI-based programs in dental practices is progressing. There is a need to evaluate the development, efficacy, and limitations of AI models tailored for prosthodontic applications. Few studies have investigated how AI-based applications are used in prosthodontics.⁷ The purpose of this study was to systematically review the performance of DL techniques for prosthodontic procedures. The research hypothesis was that DL models can improve the clinical outcomes and properties of tooth-supported fixed dental prostheses (FDPs) and removable dentures, improve prosthetic design and treatment planning, enhance color matching, facilitate automated landmark detection for RPD design, and predict facial changes after CD treatment.

MATERIAL AND METHODS

The population, intervention, comparison, and outcome (PICO) question established for developing the Boolean search of the present study was "In prosthodontics (population), which deep learning (intervention) applications have been employed, and what is their performance (outcome) against a reference standard or standard of care (control)?" (Table 1).

For population (P), the inclusion criteria included radiographs (panoramic radiographs, bitewings, and peri-apical radiographs), intraoral photographs, intraoral scans, and values for prediction of properties such as color change values, and surface roughness measurements. The interventions (I) included the DL-based identification of inlays, onlays, and tooth-supported crowns and FDPs. The DL models assessed encompassed the prediction of restoration outcomes (debonding, wear resistance, fracture resistance, surface roughness, microhardness, and flexural strength) for tooth-supported restorations and removable prostheses, optimization of prosthetic design and treatment planning; accuracy of color matching in removable and fixed prostheses; automated landmark detection for RPD treatment planning; and prediction of facial deformation following CD placement. For metrics of outcome (O), studies reporting accuracy, sensitivity (also known as recall), specificity, the area under the curve (AUC), precision (also known as positive predictive value), F1-score, root mean square error (RMSE), mean square error (MSE), intersection over union (IoU) (measuring whether ground truth masks and predicted masks were similar), or incidence rate ratio (IRR), and error rate depending on the deep learning task were included. Also, in vitro studies, clinical studies, cohort studies, and randomized clinical trials were included.

Exclusion criteria included studies that did not specify details of the dataset used (including data modality), did not explain the DL model clearly, and did not evaluate any of the outlined outcomes. Studies that evaluated machine learning models were excluded. Moreover, implant guide studies and studies about education and learning were excluded. Conference abstracts that did not allow sufficient data extraction were also excluded.

The following databases were searched electronically in August 2023 and updated in December 2024: Medline (via PubMed), Scopus, Embase, Web of Science, ArXiv, IEEE, and Google Scholar. No limitations were imposed on language. The time limit was not considered. The search query and search results are described in Table 1. Also, references to included studies were screened to identify any potential studies. Additionally, a manual search was completed.

A software program (Mendeley, v 2.89.0.; Elsevier Inc) was used to manage and assemble the included articles. Duplicated records were eliminated. Based on the defined

Table 1. Specific search query designed for each database

Data Base	Search Query	Results
PubMed	("prosthodontics"[MeSH Terms] OR "tooth preparation, prosthodontic"[MeSH Terms] OR "Dental prosthesis"[MeSH Terms] OR "dentures"[MeSH Terms] OR "denture, complete"[MeSH Terms] OR "denture, partial"[MeSH Terms] OR "denture, partial, fixed"[MeSH Terms] OR "denture, partial, removable"[MeSH Terms] OR "denture, partial, fixed, resin bonded"[MeSH Terms] OR "denture precision attachment"[MeSH Terms] OR "denture, overlay"[MeSH Terms] OR "Overdenture"[Text Word] OR "smile design"[Text Word] OR "crowns"[MeSH Terms] OR "tooth crown"[MeSH Terms] OR "dental bridge"[Text Word] OR "dental implants"[MeSH Terms] OR "dental veneers"[MeSH Terms] OR "full mouth reconstruction"[Text Word] OR "full mouth rehabilitation"[Text Word] OR "finishing line"[Text Word] OR "tooth, artificial"[MeSH Terms] OR "dental marginal adaptation"[MeSH Terms] OR "fixed prosthodontics"[Text Word] OR "post and core technique"[MeSH Terms] OR "color selection"[Text Word] OR "shade selection"[Text Word] OR "intraoral scanner"[Text Word] OR "digital impression"[Text Word] OR "intraoral digital scan"[Text Word]) AND ("artificial intelligence"[MeSH Terms] OR "computational intelligence"[Text Word] OR "machine intelligence"[Text Word] OR "computer reasoning"[Text Word] OR "AI-based"[Text Word] OR "computer vision systems"[Text Word] OR "knowledge acquisition"[Text Word] OR "deep learning"[MeSH Terms] OR "hierarchical learning"[Text Word] OR "machine learning"[MeSH Terms] OR "unsupervised machine learning"[MeSH Terms] OR "supervised machine learning"[MeSH Terms] OR "knowledge representation"[Text Word] OR "expert systems"[MeSH Terms] OR "fuzzy logic"[MeSH Terms] OR "natural language processing"[MeSH Terms] OR "neural networks, computer"[MeSH Terms])	643
Scopus	("artificial intelligence" OR "deep learning" OR "machine learning" OR "neural network") AND ("prosthodontics" OR "Dental prosthesis" OR "denture" OR "tooth crown" OR "dental bridge" OR "dental implants" OR "dental veneers" OR "finishing line" OR "fixed prosthodontics" OR "color selection" OR "intraoral scanner" OR "overlay")	661
Embase	("prosthodontics"/exp OR 'prosthodontics' OR 'prosthesis'/exp OR 'prosthesis' OR 'denture'/exp OR 'denture' OR 'crown'/exp OR 'crown') AND ('bridge'/exp OR 'bridge') OR 'implants'/exp OR 'implants' OR 'veneers' OR 'finishing line' OR 'fixed prosthodontics' OR 'color selection' OR 'intraoral scanner'/exp OR 'intraoral scanner' OR 'overlay') AND ('artificial intelligence'/exp OR 'artificial intelligence' OR 'deep learning'/exp OR 'deep learning' OR 'machine learning'/exp OR 'machine learning' OR 'neural network'/exp OR 'neural network') AND ('tooth'/exp OR 'tooth' OR 'teeth'/exp OR 'teeth' OR 'dental'/exp OR 'dental' OR 'dentistry'/exp OR 'dentistry')	1202
Web of Science	("artificial intelligence" OR "deep learning" OR "machine learning" OR "neural network") AND ("prosthodontics" OR "Dental prosthesis" OR "denture" OR "tooth crown" OR "dental bridge" OR "dental implants" OR "dental veneers" OR "finishing line" OR "fixed prosthodontics" OR "color selection" OR "intraoral scanner" OR "overlay")	637
ArXiv	AND all="prosthodontics" OR "prosthesis" OR "denture" OR "crown" OR "bridge" OR "implants" OR "veneers" OR "finishing line" OR "fixed prosthodontics" OR "color selection" OR "intraoral scanner" OR "overlay"; AND all="tooth" OR "teeth" OR "dental" OR "dentistry"	52
IEEE	("All Metadata": "prosthodontics" OR "All Metadata": "prosthesis" OR "All Metadata": "denture" OR "All Metadata": "crown" OR "All Metadata": "implants" OR "All Metadata": "veneers" OR "All Metadata": "finishing line" OR "All Metadata": "fixed prosthodontics" OR "All Metadata": "color selection" OR "All Metadata": "intraoral scanner" OR "All Metadata": "overlay") AND ("All Metadata": "tooth" OR "All Metadata": "teeth" OR "All Metadata": "dental" OR "All Metadata": "dentistry") AND ("All Metadata": "artificial intelligence" OR "All Metadata": "deep learning" OR "All Metadata": "machine learning" OR "All Metadata": "neural network")	92
Google Scholar	allintitle:("artificial intelligence" OR "deep learning" OR "machine learning" OR "neural network") AND (Prosthodontics OR prosthesis)	72

inclusion and exclusion criteria, 3 independent reviewers (K.K., P.M., T.S.) screened the title and abstracts. Insufficient information in the abstract led to the consideration of the full text. Any disagreements were resolved by consulting a fourth reviewer (R.R.). Data were extracted independently by 3 reviewers (K.K., P.M., T.S.). A consensus was reached with other reviewers in case of disagreement. Another reviewer (R.R.) also rechecked the final tables. The following items were extracted from the included studies: bibliographic details (name of authors and the year of publication), aim, prosthetic type, data modality, sample size (training, validation, and testing dataset, if available), machine learning task, labeling procedure, preprocessing procedure, augmentation, model structure, performance metrics, and outcome.

The bias assessment was conducted using the JBI Critical Appraisal Checklist for Quasi-Experimental

Studies⁸ by 3 evaluators (K.K., P.M., T.S.), achieving a 90% agreement rate. Any reviewer discrepancies were settled through a consensus process involving a fourth reviewer (R.R.). The evaluation criteria consisted of several critical questions designed to assess the rigor of the studies. These questions are specified in Table 2.⁹

RESULTS

Fig. 1 illustrates the selection process, and Tables 3 and 4 provide a summary of the key outcomes collected from the included studies. Of the 3359 studies reviewed, 31 met the inclusion criteria. The characteristics of the studies included in this review were divided into 2 main categories: studies focused on the DL model and its training process and investigations evaluating the

Table 2. Questions for risk of bias assessments, based on Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Quasi-Experimental Studies⁸

1	Did the study clearly identify and measure the variables demonstrating cause and effect relationships?
2	Was there adequate comparison between participants to ensure group equivalence at baseline?
3	Did all groups, with the exception of the intervention group, receive equivalent treatment or care throughout the study?
4	Was a proper control group utilized as part of the experimental design?
5	Were measurements conducted both before and after the intervention to establish baseline and change data?
6	Was the follow-up process thorough, with minimal loss to follow-up and appropriate handling of missing data?
7	Were the methods used to measure outcomes consistent across all comparison groups?
8	Were the data analyzed using appropriate statistical methods that align with the study design and research questions?
9	Were the outcome measurements consistent, accurate, and appropriately selected to address the research objectives?

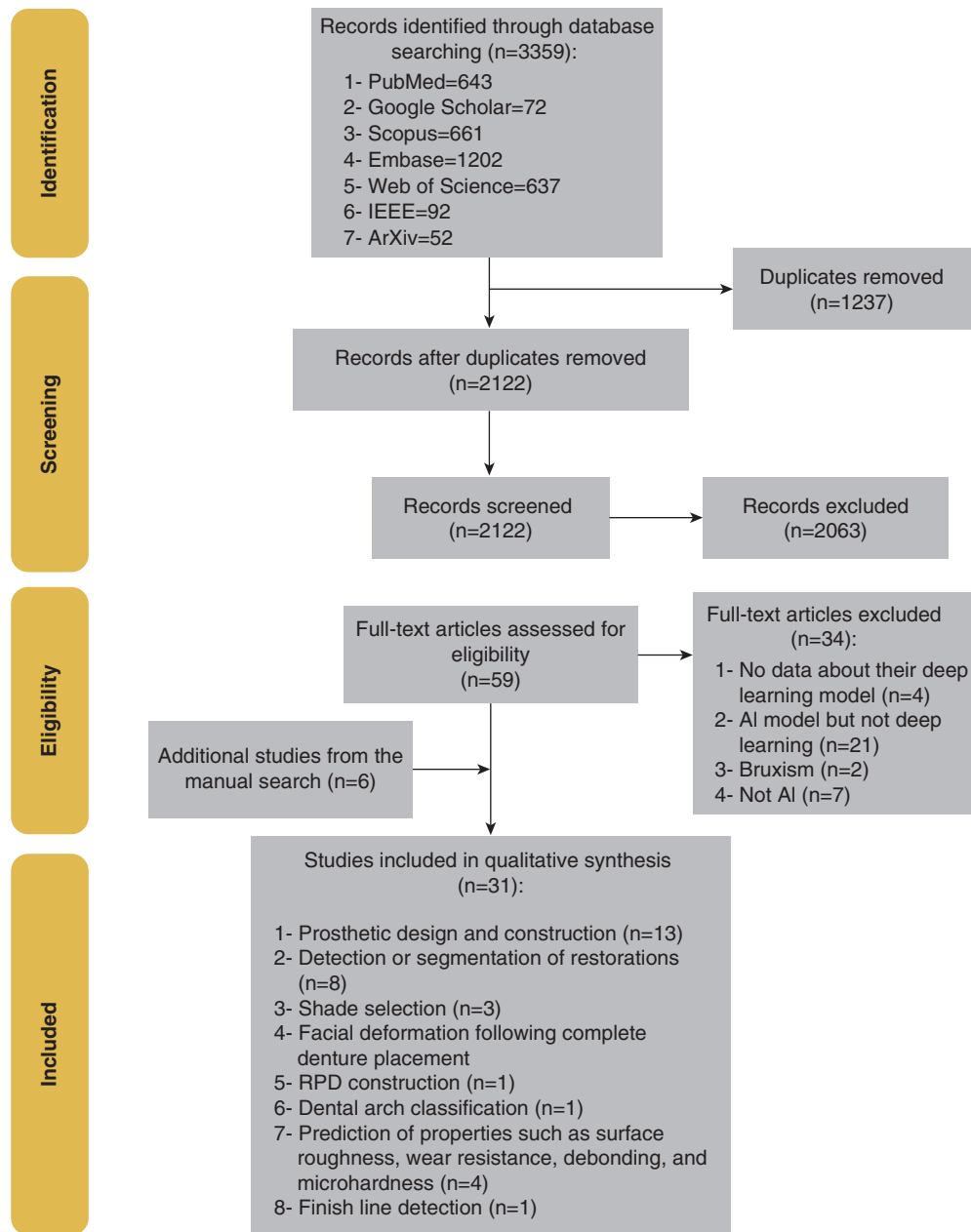


Figure 1. PRISMA flowchart for selection of studies.

prosthodontic applications, including objectives, data modalities, and types of prosthodontics involved.

The included studies varied in the use of DL models developed for different tasks (Fig. 2). The most used model was the Convolutional Neural Network (CNN) in 13 studies,^{4,5,10–20} followed by Generative Adversarial Networks (GAN) in 7 studies,^{4,18,21–25} Artificial Neural Networks (ANN)^{13,26–30} and Neural Networks (NN)^{31–36} in 6 studies, and Natural Language Processing (NLP) in 1 study.³⁷

Regarding the tasks performed by DL, the studies showed a variety of restorative applications. The most frequent task was generation (when a model creates new data

instances such as text, images, or other complex data structures that are similar to the training data but not direct copies) (n=11),^{5,10,18,21–26,33,38} followed by regression (predicting a continuous output variable based on one or more input variables) (n=10),^{17,20,27,29,30,32,34–37} object detection (n=8),^{10,12–16,26,28} segmentation (n=8),^{10,11,14,19,21,22,33,38} and classification (n=4).^{4,13,14,19,31}

Concerning the objectives of the reviewed studies, the majority focused on prosthetic design and fabrication (n=13).^{5,10,18,20,21,23–26,31,33,37,38} Other analyzed areas included the detection or segmentation of restorations (n=8),^{11–16,28,38} shade selection (n=3),^{34–36} facial deformation following CD placement (n=1),³⁰

Table 3. An overview of each study's main properties

Author, Year, Ref	Aim and Prosthetic Type	Data Modality, Sample Size (Train/ Validation/ Test)	Machine Learning Task, Labeling Procedure
Choi et al, 2024 ¹⁹	Finish line detection	IOS, 180	Segmentation, classification
Cho et al, 2024 ²⁶	Dental crown	IOS,	Object detection, generation
Li et al, 2024 ¹⁰	Dental crown	30	Segmentation, object detection, generation
Chen et al, 2024 ³¹	Prosthetic design	IOS,	Classification
Chen et al, 2024 ³¹	Complete denture	NA	Segmentation, generation
Chau et al, 2023 ²¹	Prosthetic design	IOS,	Labeling by two operators
Chau et al, 2023 ²¹	Restoration	90 (80/10/-)	Prediction
Feng et al, 2023 ²⁰	Dental crown	IOS,	Prediction
Feng et al, 2023 ²⁰	Crown design	250(200/-/50)	Segmentation, generation
Grymak A. et al, 2023 ³²	Maxillary anterior crowns	600(480/-/120)	Labeling by two operators
Grymak A. et al, 2023 ³²	Prediction of wear resistance of denture teeth materials	In vitro study data (wear measurement)	Prediction
Gardiyanoglu, E., et al, 2023 ¹¹	Denture	60 (10/20/ 30)	Segmentation, labeling by two maxillofacial radiologists with manual drawing semantic segmentation technique
Gardiyanoglu, E., et al, 2023 ¹¹	Automatic segmentation of various objects on panoramic	Panoramic	Segmentation, labeling by two maxillofacial radiologists with manual drawing semantic segmentation technique
Gardiyanoglu, E., et al, 2023 ¹¹	Teeth, crown-fixed partial dentures, implants, composite-amalgam fillings, caries, residual roots, RCTs	8138 (80%/-/20%)	Segmentation, labeling by two maxillofacial radiologists with manual drawing semantic segmentation technique
Hosseinihanesh, G., et al, 2023 ³³	Prosthetic design	IOS	Segmentation, generation, labeling: Technicians into 4 tooth types: incisor, canine, premolar, molar
Shen et al, 2023 ²²	Dental crown	125 (90/10/25)	Segmentation, generation
Shen et al, 2023 ²²	To generate RPD clasps	IOS	Segmentation, generation
Ding, H., et al, 2023 ³⁸	RPD clasps	39 (30/9/-) A four-fold cross-validation was used	Segmentation, generation
Ding, H., et al, 2023 ³⁸	Prosthetic design	IOS	Segmentation, generation
Dental crown	Dental crown	612(600/-/12)	Generation, labeling by 2 clinicians
Farook, T., H., et al, 2023 ⁵	Prosthetic design	IOS	Generation
Farook, T., H., et al, 2023 ⁵	Partial dental crowns (inlay, onlay)	30(20/10/-)	Object detection, labeling by 3 experienced clinicians (70 landmarks were annotated for each IOS using 3D slicer)
Cho et al, 2023 ²³	Prosthetic design	IOS	Object detection, classification, labeling by an Oral and Maxillofacial radiologist
Dental crown	Dental crown	30	Object detection, classification, labeling by an Oral and Maxillofacial radiologist
Automated dental crown detection	Automated dental crown detection	IOS 4	Object detection, labeling by 3 experienced clinicians (70 landmarks were annotated for each IOS using 3D slicer)
Baquero, B, 2023 ¹²	Dental crown	05 (five-fold cross validation)	Object detection, labeling by 3 experienced clinicians (70 landmarks were annotated for each IOS using 3D slicer)
Çelik, B., et al, 2022 ¹³	Automatic detection of dental restoration	Panoramic	Object detection, classification, labeling by an Oral and Maxillofacial radiologist
Dental restorations (Filling and single-unit fixed crowns), multiunit fixed crowns, implant	Dental restorations (Filling and single-unit fixed crowns), multiunit fixed crowns, implant	684 (80/10/10)	Object detection, classification, labeling by specialist dentist
Detection of dental materials	Detection of dental materials	Panoramic	Object detection, classification, labeling by specialist dentist
Dental restoration, fixed partial denture and tooth crown regions	Dental restoration, fixed partial denture and tooth crown regions	487 (80%/-/20%)	Object detection, classification, labeling by specialist dentist
Engels, P, et al, 2022 ¹⁵	Automated detection and categorization of posterior restorations	Five-fold cross-validation	Object detection, classification, labeling by specialist dentist
Engels, P, et al, 2022 ¹⁵	Composite resin, cement, amalgam, gold and ceramic restorations	IOP	Object detection by 3 dentists and then experience dentist
Dijmarescu et al, 2021 ²⁷	Surface roughness prediction of dry end-milling Co-Cr alloy; Co-Cr alloy	3522 (2814/-/708)	Prediction, labeling by the experimental data using the Sub-VIS
Takahashi et al, 2021 ²⁸	Recognition of prostheses and restorations	Surface roughness (Ra) measurements	Object detection, labeling manual annotation in 8 categories (CVC, GCMC, RFMC, PFMC, CC, Br, RFMCBr, PFMCBr) and 3 restorations (CR, Mh, Gin)
Tian et al, 2022 ²⁴	Prosthetic design and construction Crown	NA	Generation
Tian et al, 2021 ²⁵	Prosthetic design and construction Inlay and onlay	IOP	Generation
Tian et al, 2021 ²⁵	Prosthetic design and construction Inlay and onlay	1904 (867/-/380)	Generation
Tian et al, 2021 ²⁵	Prosthetic design and construction Inlay and onlay	IOS	Generation
Tian et al, 2021 ²⁵	Prosthetic design and construction Inlay and onlay	780 (700/-/80)	Generation
Tian et al, 2021 ²⁵	Prosthetic design and construction Inlay and onlay	IOS	Generation
Tian et al, 2021 ²⁵	Prosthetic design and construction Inlay and onlay	830 (710/40/80)	Generation

Table 3 (Continued)

Author, Year, Ref	Aim and Prosthetic Type	Data Modality, Sample Size (Train/ Validation/ Test)	Machine Learning Task, Labeling Procedure
Takahashi et al, 2021 ⁴	Dental arch classification RPD	IOP 1184 (1016/-/ 168)	Classification, labeled to edentulous, intact dentition, with posterior tooth loss, and with bounded edentulous space
Gurses, A., et al, 2020 ¹⁶	Automatic detection of dental work and type of restorations Raw teeth Implants, RCT, amalgam or composite resin restorations, orthodontic appliances	Panoramic 3013 (2113/-/900)	Object detection, labeling by an expert
Ueki et al, 2020 ³⁴	Shade selection In vitro data (Shade guide)	IOP 62 (32/8/22)	Prediction, labeling by An experienced dental laboratory technician (21 color shade guide)
Yamaguchi et al, 2019 ¹⁷	Predicting debonding probability CAD-CAM composite resin crowns	2D images captured from 3D stereolithography models of die scanned using 3D oral scanner 8640 (6480/-/2160)	Prediction, labeling as debonding or trouble-free treatments
Deniz ST. et al, 2019 ²⁹	Prediction of microhardness and surface roughness Denture	In vitro data 200	Regression
Wei et al, 2018 ³⁵	Shade selection Metal-ceramic disk-shaped specimens	In vitro data (Color coordinates (CIE L*a*b* values) were measured with a spectroradiometer as the input) 43 (39/10/4)	Prediction
Hwang et al, 2018 ¹⁸	Prosthetic design Dental crown	IOS 3313 (1500/ 1570/243)	Generation
Chen, Q., et al, 2016 ³⁷	Prosthetic design RPD	Clinical examination data (ontology database) 104 (99/-/5)	Prediction, Preparing a knowledge graph based on the oral condition examination and RPD design principles
Li et al, 2015 ³⁶	Shade selection Ceramic restorations	In vitro data (Shade measurement (L*, a*, and b*) of the specimen with a dental Spectrophotometer) 119 (75%/-/25%) 3D face scan 48(43/-/5)	Prediction
Cheng et al, 2015 ³⁰	Facial deformation prediction after complete denture prosthesis Complete denture prosthesis		

Table 4. Overview of each study's findings

Author, Year, Ref	Preprocessing Augmentation	Model (Algorithm Architecture)	Performance Metric	Outcome
Choi et al, 2024 ¹⁹	NA	CNN	Hausdorff distance threshold (desktop scan)	0.366
			Hausdorff distance threshold (intraoral scan)	0.566
			Chamfer distance threshold (desktop scan)	0.026
			Chamfer distance threshold (intraoral scan)	0.100
Cho et al, 2024 ²⁶	NA	Dentbird Crown; Imageworks Inc; Automate; 3Shape	RMSE	Internal fit (Automate): 83.1 ±13.1 Internal fit (Dentbird): 59.4 ±12.1
Li et al 2024 ¹⁰	NA	SegNet (DGCNN, PointCNN, PointNet, PointNet++, LWN)	Accuracy	Major connector: 98.54 ±0.58%
			IoU	Retentive mesh: 97.73 ±0.92% Major connector: 87.42 ±5.48% Retentive mesh: 70.42 ±7.95%
Chen et al 2024 ³¹	Alignment	NN (ToothDIT)	RMSE	Corner defect: 0.112 Incisive defect: 0.125 Medium defect: 0.127
Chau et al 2023. ²¹	Duplicating dataset, Removal of maxillary first molar from arch using MeshLab	3D GAN	IoU	60%
Feng et al 2023 ²⁰	NA	Transformer (DGCNN)	Average deviation (point cloud)	0.22
			Average deviation (crown model)	0.13
Grymak A. et al 2023 ³²	Resampling, temporal smoother	Long short-term memory recurrent neural network	Accuracy	96.4%
			RMSE	6.23–88.56
			MAE	7.47–70.71
			Mean absolute percentage error	12.43%–23.02%
Gardiyanoglu, E., et al, 2023 ¹¹	Resizing	U-Net	Accuracy	Teeth:95% Caries:99% Restorations:99% Crown-fixed partial denture restorations:99% Implants:99% RCTs:99% Residual Roots:99%
Hosseinimanes, G., et al, 2023 ³³	Preprocessing: semantically segmentation of teeth into 14 classes (number of teeth in each jaw) using MeshSegNet method, extracting margin line from prepared tooth; augmentation: rotation, translation, scaling	Transformer encoder-decoder architecture	With margin line information	Average margin distance (µm): 372.2782 ±184.4864
			Without margin line information	Average margin distance (µm): 2105.3221 ±783.3138
Shen et al, 2023 ²²	NA	PCN, Pre-trained PCN, GRNet, Pre-trained GRNet, PoinTr, Pre-trained PoinTr, 3D-RecGAN, IF-Net, TransSDFNet CNN	F-Score@1% (PCN)	23.89% ±8.31%
			F-Score@1% (Pre-trained PCN)	20.98% ±9.92%
			F-Score@1% (GRNet)	29.50% ±4.54%
			F-Score@1% (Pre-trained GRNet)	51.98% ±5.12%
			F-Score@1% (PoinTr)	60.31% ±7.68%
			F-Score@1% (Pre-trained PoinTr)	42.01% ±14.00%
			F-Score@1% (3D-RecGAN)	69.74% ±6.06%
			F-Score@1% (IF-Net)	73.91% ±4.50%
			F-Score@1% (TransSDFNet CNN)	82.88% ±3.43%
Ding, H., et al, 2023 ³⁸	Manual segmentation by Meshlab software program	3D-Deep Convolutional Generative Adversarial Network (3D-DCGAN)	RMSE	0.3611
Farook, T., H., et al, 2023 ⁵	Manual trimming	3D-CNN	Accuracy	60%
			Sensitivity	100%
			Precision	83%
Cho et al, 2023 ²³	NA	GAN	RMSE	Difference in occlusal morphology: 54.1 ±25.5 Internal fit to prepared abutment: 55.4 ±17.1

Table 4 (Continued)

Author, Year, Ref	Preprocessing Augmentation	Model (Algorithm Architecture)	Performance Metric	Outcome
Baquero, B., 2023 ¹²	Tooth Segmentation and numbering using DentalModelSeg and Universal Labeling tool in 3D slicer	U-Net (ALIIO5 CNN algorithm)	Average distance error	0.43 ±0.28mm (lower occlusal landmarks) 0.45 ±0.28mm (upper occlusal landmarks) 0.62 ±0.28mm (lower and upper cervical landmarks) (0.75–0.97) for all models
Çelik, B., et al, 2022 ¹³	Converting to PNG, resizing	R-CNN, Faster R-CNN, SSD, Yolov3, RetinaNet, ResNet–50, ResNet–101, Xception–101, VGG16, DarkNet53	mAP Area under Precision-recall curve	0.95
Imak, A., et al, 2022 ¹⁴	Denoising and contrast enhancement	Faster R-CNN (AlexNet and RPN), Graph cut, SVM, K-NN, decision tree, Naïve Bayes	Accuracy (SVM) Accuracy (K-NN) Accuracy (DT) Accuracy (NB) AUC (SVM)	92.8% 89.9% 89.7% 89.5% 0.92
Engels, P., et al, 2022 ¹⁵	Cropping, rotation, multiplying, transformations (cropping, deletion, affine transformation, perspective transformation, changes in brightness, contrast and saturation) and resizing	CNN (ResNeXt–101–32 × 8d)	Accuracy	Unrestored teeth: 94.9% Composites: 92.9% Cements: 98.3% Amalgam: 99.2% Gold: 99.4% Ceramic: 97.8%
			Sensitivity	Unrestored teeth: 93.9% Composites: 88.6% Cements: 84.6% Amalgam: 100% Gold: 93.8% Ceramic: 86.4%
			Specificity	Unrestored teeth: 95.2% Composites: 95.2% Cements: 99.4% Amalgam: 99% Gold: 100% Ceramic: 99.4%
			AUC	Unrestored teeth: 99.2% Composites: 98.7% Cements: 98.7% Amalgam res: 100% Gold: 99.7% Ceramic: 99.3%
Dijmărescu, M., R., et al 2021 ²⁷	Data normalization, image processing software	ANN	Prediction error R^2	0.5% 99%
			Prediction error R^2	1.19% 99%
Takahashi et al, 2021 ²⁸	Preprocessing: resize	YOLO v3	Average precision	CMC:93% GCMC:90% RFMC:78% PFMC:78% CC:59% RFMCBr:81% PFMCBr:77% Br:77% Min:92% GIn:90% CR:61% 80% 76%
Tian et al, 2022 ²⁴	Preprocessing: normalization using bounding box, virtual rotating sphere technology	Pix2pix, Pix2pixHD, PAN, GFC, Dental-GAN, Stage-I GAN, Stage-I GroNet_OF, DCPR-GAN	MAP mIoU RMSE (Pix2pix) RMSE (Pix2pixHD) RMSE (PAN) RMSE (GFC) RMSE (Dental-GAN) RMSE (Stage-I GAN) RMSE (Stage-I GroNet_OF) RMSE (DCPR-GAN)	0.079 ±0.007 0.073 ±0.005 0.077 ±0.008 0.075 ±0.007 0.072 ±0.012 0.071 ±0.009 0.072 ±0.008 0.068 ±0.008
Tian et al, 2021 ²⁵	NA	Pix2pix, PAN, GFC, Dental-GAN, DAIS	RMSE (Pix2pix) RMSE (PAN) RMSE (GFC) RMSE (Dental-GAN) RMSE (DAIS)	10.11 8.36 7.42 7.53 7.15

Table 4 (Continued)

Author, Year, Ref	Preprocessing Augmentation	Model (Algorithm Architecture)	Performance Metric	Outcome
Takahashi et al, 2021 ⁴	Preprocessing: resize The ImageNet pre-trained model of 152-layer residual network model (ResNet152) with fine-tuning used for preprocessing	CNN	Diagnostic accuracy Precision Recall F-measure	Max: 99.5% Mand: 99.7% Max: 25% Mand: 25% Total: 25% Max: 100% Max: 40% Mand: 40%
Gurses, A., et al, 2020 ¹⁶	Normalizing, resizing, cropping	CNN (DenseNet, GoogleNet, ResNet)	Accuracy (DenseNet) Accuracy (GoogleNet) Accuracy (ResNet)	94% 89% 93%
Ueki et al, 2020 ³⁴	Augmentation performed by extracting 1024 data from each image for training and testing	Neural network	Estimation accuracy of confidence values	The estimation accuracy was 27%, 55% and 68% for the first, second and third candidates respectively.
Yamaguchi et al, 2019 ¹⁷	Preprocessing: crop, resize, rotate; Augmentation: rotation, width and height shifting, zooming, shearing, and horizontal flip	CNN	Accuracy Precision Recall F-measure AUC	98.5% 97.0% 100% 0.985 0.998
Deniz ST. et al 2019 ²⁹	NA	BPANN+LM	MSE	7.65 (microhardness) and 4 (surface roughness)
Wei et al, 2018 ³⁵	Preprocessing: Normalization of inputs and outputs to constraint interval [-1, 1]	BPNN + GA + LM	Color difference measurement (between fabricated specimen and natural tooth)	1.89 ±0.75 Computer color matching based on the new BPNN model had significantly less color difference than the conventional visual method
Hwang, J., J., et al, 2018 ¹⁸	NA	GAN U-Net	IOU Precision Recall F-Measure of contours RMSE	91.5–92.2% 93%–94.4% 94.4%–95.4% 93.8–94.9% 0.06–0.07%
Chen, Q., et al, 2016 ³⁷	NA	NLP	Precision AUC-ROC MAP	67% 96% 61%
Li et al, 2015 ³⁶	Preprocessing: Normalization Augmentation: crossover and mutation (GA steps)	BPNN + GA + LM, and BNPP	MSE (BPNN + GA + LM)	0.030
Cheng et al 2015 ³⁰	NA	BP ANN	MSE (BNPP) Error rate	0.033 39.24%

Prosthesis: Br, bridge; CAD-CAM, computer-aided design and computer-aided manufacturing; CC, ceramic crown; CMC, silver-colored complete metal crown; CR, composite resin filling; GCMC, gold-colored complete metal crown; Gln, gold-colored metal inlay restoration; Mln, silver-colored metal inlay restoration; PFMC, porcelain fused to metal crown; RFMC, resin facing metal crown; RPD, removable partial denture.

Model: CNN, convolutional neural network; DAIS, deep adversarial-driven dental inlay restoration framework; DCPR-GAN, dental crown prosthesis restoration generative adversarial network; Dental-GAN, dental occlusal surface generator network; GAN, generative adversarial network; GFC, generative face completion network; NLP, natural language processing; PAN, perceptual adversarial network; R-CNN, regions with convolutional neural network; Stage-I GroNet_OF, first stage groove parsing network occlusal fingerprint; TranSDFNet, transformer-based truncated signed distance fields u-shape network; YOLO, you only look once.

Metric: AR, average recall; AUC, area under the curve; CD-L2, chamfer-L2 distance; DSC, Dice similarity coefficient; EMD, earth mover's distance; FN, false negative; FP, false positive; FSIM, feature similarity index measure; HD, Hausdorff distance; LR, likelihood ratio; MAE, mean absolute error; MAP (mAP), mean average of precision; mIoU, mean intersection over union; MSE, mean square error; MS-SSIM, multi-scale structural similarity index measure; NDCG, normalized discounted cumulative gain; NPV, negative predictive value; PPV, positive predictive value; PSNR, peak signal noise ratio; R², the coefficient of determination; RMSE, root mean square error; SSIM, structural similarity index measure; TN, true negative; TP, true positive.

Other: AI, artificial intelligence; IOPs, intraoral photographs; IOSs, intraoral scans; Mand., mandible; Max., maxilla; NA, not available; PAN, panoramic radiographs.

RPD design (n=1),²² dental arch classification (n=1),⁴ finish line of tooth preparation detection (n=1), and prediction of mechanical properties of restorations, such as surface roughness,²⁷ wear resistance,³² debonding,¹⁷ and microhardness.²⁹

In the studies reviewed, the types of prostheses evaluated most frequently among the reviewed studies were crowns (tooth-supported and implant-supported) (n=14),^{11–13,17–21,23,24,26,28,33,38} followed by restorations (such as amalgam, composite resin, and gold

(n=8),^{11,13–16,28,31,36} restoration materials (n=4),^{14,15,27,35} CDs (n=3),^{10,29,30} inlay or onlay restorations (n=3),^{5,25,28} RPDs (n=2),^{4,37} FDPs (n=1),²⁸ denture teeth (n=1),³² and RPD clasps (n=1).²²

Concerning the sample size, 2 studies^{10,27} did not specify the number of specimens included. The included studies trained the DL model on different dental imaging modalities, and digital scans recorded by using intraoral scanners (IOSs) were the most commonly used (n=16).^{5,10,12,17–26,31,33,38} This was followed by in vitro

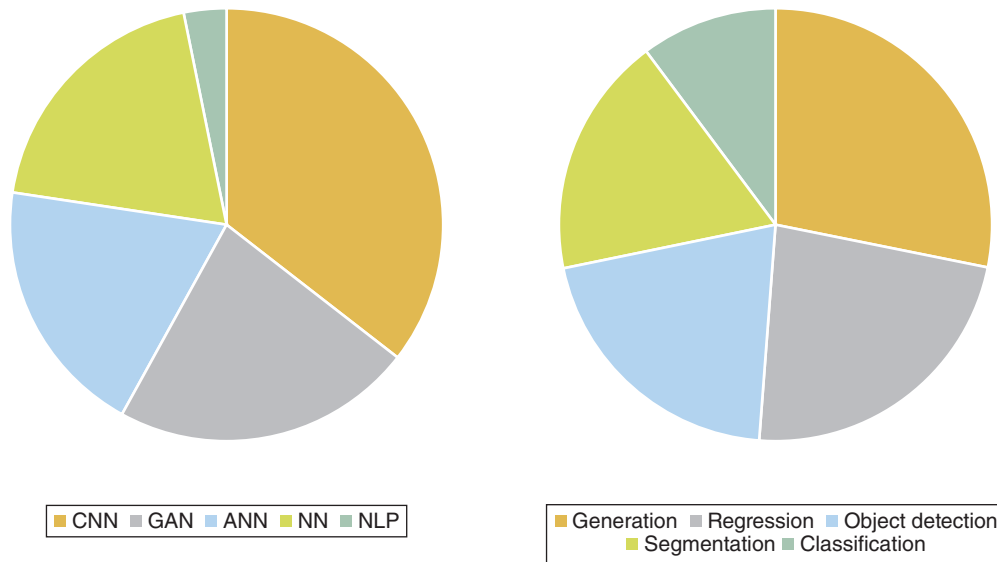


Figure 2. Distribution of included studies based on deep learning models and tasks.

data measurements (wear measurements, surface roughness, and color values) ($n=7$),^{17,27,29,32,34–36} panoramic radiographs,^{11,13,14,16} and intraoral photographs,^{4,15,28,34} each used in 4 studies. Additionally, clinical data³⁷ and 3D facial scans³⁰ were used in 1 study. Additionally, 8 studies^{5,22,23,26,30,31,34} reported a sample size of less than 100 specimens.

A JBI checklist was used to evaluate the studies. However, in the present systematic review, specific questions of the JBI checklist were not applicable and, thus, not considered for the included studies (questions 2, 3, 5, 6, and 7). Questions 1, 4, 8, and 9 were applied to the included studies. Fig. 3 displays the outcomes from the JBI Checklist for Quasi-Experimental Studies applied to the included studies. Ten studies were categorized as low risk of bias across all checklist questions.

Different metrics were used across the studies to report the analyzed outcomes. The most used metrics were RMSE and MSE, which appeared in 10 studies.^{18,23–26,29,31,32,36,38} Accuracy was the second most

frequently used metric, reported in 9 studies. Accuracy ranged from 60% to 99%, sensitivity spanned from 88.6% to 100%, IoU varied from 60% to 90%, and precision ranged from 25% to 93%.^{8,9,15,20–23,36} Farrok et al⁵ reported an accuracy of 60% for an onlay and other prosthetic designs using a CNN model.⁹ Gardiyanoglu et al¹¹ achieved a 99% accuracy rate in detecting restorations, tooth-supported crowns, and FPDs, and implants by using a U-Net model. Engels et al¹⁵ also reported a 99.4% accuracy in identifying gold restorations using a CNN model. Similar high accuracy has been observed in the classification of RPDs, with Takahashi et al⁴ achieving a 99.7% accuracy for classifying the maxillary arch using a CNN model.⁸ Grymak et al³² reported a 96.4% accuracy in predicting wear resistance with a neural network model. Additionally, Yamaguchi et al¹⁷ achieved a 98.5% accuracy in predicting the likelihood of debonding for CAD-CAM composite resin crowns using a CNN model. Also, Li et al³⁶ observed an MSE of 0.033 using a BPNN model to predict shade selection in ceramic restorations. The RMSE is a gauge for assessing a model's error in quantitative predictions. It calculates the square root of the average squared differences between predicted and actual values. A lower root mean square error (RMSE) suggests that the model's predictions are more accurate, indicating higher predictive accuracy.³⁶ Additionally, Ding et al³⁸ reported an RMSE of 0.3611 in their study on the morphology and mechanical performance of dental crown design using a GAN model.

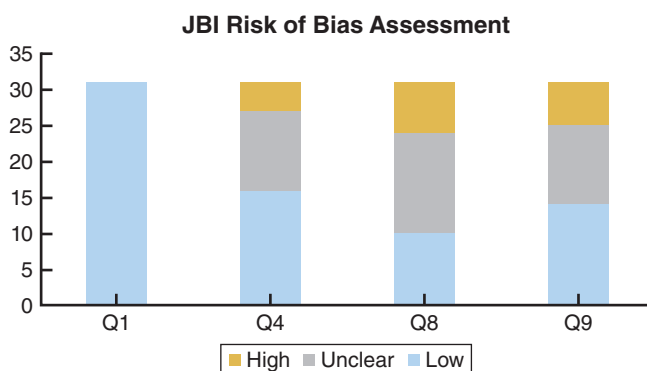


Figure 3. Assessment of bias risk across various domains of included studies.

DISCUSSION

This review systematically analyzed the applications of DL in prosthodontics that involved several capacities

such as predictive analytics, automated diagnostic processes, and tools for identification and classification (such as RPD and restorations). Specifically, DL models were used in areas such as CAD-CAM systems for designing dental restorations and detecting the finish line of tooth preparations, designing implant and tooth-supported crowns, and analyzing orofacial anatomy within the broader scope of prosthodontics.

The examined studies showed variability in deep learning model training, machine learning applications, and prosthodontic modalities. GANs predominated among the implemented models, featuring generators creating data samples and discriminators evaluating authenticity. These components train competitively, enhancing the generator's ability to produce realistic outputs—advantageous for limited medical imaging datasets. However, GANs present challenges including complex complexity, resource intensity, and training difficulties.³⁹

The widespread use of GANs in the reviewed studies is likely attributable to their strengths in several key areas relevant to dentistry. GANs excel at generating realistic images of dental restorations, a capability valuable for treatment planning and patient communication.^{24,38} Their capacity to enhance the quality of radiographic images is also significant, improving diagnostic accuracy.^{24,38} Furthermore, GANs can be used to augment existing datasets and create 3D models of dental structures, both of which are crucial for developing and refining AI-driven tools.^{24,38} Finally, the potential of GANs to assist in treatment planning and simulation by allowing for the visualization of potential posttreatment outcomes adds to their utility and explains their prominent role in this area of research.²⁴

The training process for DL models varied significantly among the reviewed studies depending on the specific objectives, particularly in prosthodontics. For instance, when using DL to design RPDs, the system must initially gather comprehensive intraoral data such as the positions of missing teeth, the condition of the remaining teeth, and their occlusal relationships. In contrast, the approach for detecting dental restorations in radiographic images does not require as detailed an initial data collection level.^{4,22,37}

In prosthodontics, the Kennedy classification provides a standardized framework for classifying RPDs; however, a lack of consensus regarding optimal RPD design complicates the establishment of robust reference standards for research⁴ while DL models have not been widely applied to predict restoration outcomes such as debonding, wear resistance, fracture resistance, surface roughness, and microhardness. The reviewed studies demonstrate high prediction accuracy, suggesting significant future potential.

Inconsistencies in reported metrics across studies, despite similar objectives and tasks, highlight the absence of standardized reporting guidelines for AI models in prosthodontic research.

The included studies exhibited several limitations that impacted their findings and the applicability of their models. The potential for generalizing these models was not evaluated across external datasets (external validation). This limitation is particularly critical when the data originate solely from a single center, lacking diversity and possibly containing hidden biases that can affect the model's performance on datasets from different sources. This underscores the importance of enhancing model generalizability. Additionally, there is an increased risk of overfitting where the model becomes excessively tailored to the training data at the expense of its broader applicability. To mitigate these issues, techniques like data augmentation, transfer learning, and integrating domain-specific knowledge are essential to bolster the robustness and reliability of AI models, particularly in the context of prosthodontic applications.² Another aspect affecting AI model performance are the specific characteristics of the dental structures being analyzed, such as tooth type and location in the dental arch. These factors can introduce complexities, including overlapping structures, varying angulations, and positions, all of which can influence the model's discriminative ability in detection or identification tasks.^{2,12,19}

A further limitation noted in the studies was the "black box" nature of DL models and their hidden layers, with failings to address the issue of model explainability. Transparency in AI is crucial, encompassing the algorithms used, the clinical data handling, and the demographic details of the dataset, such as age, sex, and nationality of the patients. For classification tasks, it should be clear why a particular class was predicted, and, for regression models, the predictors used should be well defined to ensure an understanding of the outcomes.² Additionally, there was a lack of detailed information on the validation and calibration of annotators, as well as on how disagreements among annotators were managed, which is a significant oversight. Data augmentation, particularly noise reduction and other image filters, significantly improves model performance by refining training data and enhancing accuracy.

The study also faced certain limitations that impacted the breadth and depth of the analysis. Due to the heterogeneity of data—including the use of DL models for different tasks and the reporting of various metrics—a meta-analysis was not feasible. This underscores the critical need for standardized reporting guidelines and metrics in AI-driven dental research.

Furthermore, a significant oversight in most of the included studies was the lack of comparison between the performance of deep learning models and the standard treatment typically represented by human experts. This comparison is crucial for assessing AI models' practical value and reliability in clinical settings. Lastly, the diverse methodologies and outcomes reported across the studies made it challenging to establish a consistent theme or "common tune" in the data extracted, complicating the synthesis of findings and conclusions.

CONCLUSIONS

Based on the findings of the present systematic review, the following conclusions were drawn:

1. There has been a rise in deep learning use within prosthodontics, primarily for prosthesis design and optimizing prosthesis manufacturing.
2. The deep learning models used are diverse, leading to inconsistent accuracy levels among the review studies depending on the specific task and the deep learning model.
3. Further research is necessary to create standardized evaluation methodologies, increase validation efforts, and reduce bias to maximize deep learning's potential in improving precision and efficiency.

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<https://doi.org/10.1016/j.jprosdent.2025.04.009>